Supplemental Material for "Coupled sign and magnitude memory amplifies market-order-flow noise"

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EXACT FINITE-SAMPLE REFERENCE

Let x_t be signed flow after subtracting its empirical mean within a declared (period, UTC hour-of-week) stratum. The retained observations form W complete weeks of $H = 168$ hours, and

$$Q_w = \sum_{t\in w} x_t, \qquad V_{\text{obs}} = \frac{1}{W-1}\sum_w (Q_w - \bar{Q})^2. \tag{1}$$

Every observation belongs to one week and the sum of x_t vanishes in every stratum. Hence $\bar{Q} = 0$ up to floating-point roundoff.

Expand the weekly square:

$$\sum_w Q_w^2 = \underbrace{\sum_t x_t^2}_{D} + \underbrace{\sum_w \sum_{\substack{t,t'\in w\\ t\neq t'}} x_t x_{t'}}_{C}. \tag{2}$$

A complete week contains one candle at each hour of week. It therefore contains at most one observation from any declared stratum, even when a week crosses a calendar-period boundary. Under independent within-stratum permutations, x_t and $x_{t'}$ in an off-diagonal weekly pair come from two different zero-mean strata, so $\mathbb{E}_\pi[x_{\pi(t)}x_{\pi(t')}] = 0$. Permutation also preserves D exactly. Thus

$$\mathbb{E}_\pi[V_{\text{joint}}] = \frac{D}{W-1} \equiv V_0. \tag{3}$$

This identity holds for finite samples and permutation without replacement. The deposited calculation verifies the analytic value against explicit joint permutations; the small difference between V_0 and a reported null *median* is Monte Carlo distributional asymmetry, not a physical effect.

SIGN-ORDER AND COUPLING TERMS

Write $x_t = s_t m_t$, with $s_t = \text{sign}(x_t)$ and $m_t = |x_t|$. Define stratum means

$$\bar{s}_g = \frac{1}{n_g}\sum_{t\in g} s_t, \qquad \bar{m}_g = \frac{1}{n_g}\sum_{t\in g} m_t, \qquad \widetilde{s}_t = s_t - \bar{s}_{g(t)}. \tag{4}$$

Centering the sign proxy is essential. A nonzero $\bar{s}_g$ can be offset by a contemporaneous relation between signs and magnitudes even though the signed flow itself has zero stratum mean; assigning that offset to "sign memory" would import a one-point sign–magnitude association into the sign-only term.

The homogenised sign-order contribution is

$$C_s = \sum_w \sum_{\substack{t,t'\in w\\ t\neq t'}} \widetilde{s}_t \widetilde{s}_{t'} \bar{m}_{g(t)} \bar{m}_{g(t')}. \tag{5}$$

The coupling remainder is $C_{sm} = C - C_s$. With $V_s = C_s/(W-1)$ and $V_{sm} = C_{sm}/(W-1)$, Eq. (2) gives

$$V_{\text{obs}} = V_0 + V_s + V_{sm} \tag{6}$$

to machine precision. Table I reports $A = V_{\rm obs}/V_0$, $f_s = V_s/(V_{\rm obs} - V_0)$, and $f_{sm} = V_{sm}/(V_{\rm obs} - V_0)$. No positivity constraint is imposed. A negative or greater-than-one share would remain visible and would invalidate a simple mixture interpretation.

Equation (6) is an operational allocation of an interaction, not a unique causal decomposition. Its advantages are that the reference is analytic, every term is computed from the same weekly sample, and the allocation rule is stated before interpreting the result. In particular, $V_0 + V_s$ keeps the observed diagonal D and is not $\mathrm{Var}(\widetilde{s}_t \bar{m}_{g(t)})$. Replacing $\bar{m}_g$ by another declared stratum scale would give a different exact remainder; the reported share is therefore conditional on the arithmetic-mean homogenisation, not an orthogonal or uniquely identified interaction fraction.

WHY THE EARLIER SURROGATE COMPARISON IS NOT ADDITIVE

The earlier analysis used three nulls: permuting complete x_t values, permuting signs while magnitudes remain at their observed times, and permuting magnitudes while signs remain fixed. Table IV retains those values as controls. The sign and magnitude nulls break contemporaneous sign–magnitude pairing and cannot be added as variance components.

Nor is equality of the sign and joint nulls an exact identity for the implemented algorithm. If a stratum contains n signs permuted without replacement, then for two distinct positions

$$\mathbb{E}_\pi[s_{\pi(i)} s_{\pi(j)}] = \frac{(\sum_k s_k)^2 - \sum_k s_k^2}{n(n-1)}, \tag{7}$$

which is generally nonzero. Fixed magnitudes weight these finite-population terms, and the surrogate is subsequently re-centred. Explicit enumeration of a four-week finite example gives $\mathbb{E}[V_J] = 16.6667$ and $\mathbb{E}[V_S] = 15.6667$ while satisfying Eq. (3) exactly. The empirical proximity of the two nulls is therefore a structural feature of sign randomisation, not a measured decomposition and not a literal equality.

PAIRED MOVING-BLOCK UNCERTAINTY

For each week we store the diagonal, total off-diagonal, homogenised-sign off-diagonal, and coupling contributions. A circular moving-block sample uses the same resampled week indices for every contribution. The shares are then recomputed from the paired sums; replicates with nonpositive total excess are excluded from share quantiles and their retained fraction is recorded. More than half of the 3999 replicates must remain valid or the calculation fails.

We report block lengths of 4, 8, 13, and 26 weeks. All four primary raw-flow coupling intervals remain above one half at eight weeks [Table I]. Table II shows that the conclusion is not tied to that single block choice. Stratum means are held fixed throughout; the intervals are conditional on the declared calendar partition and do not represent uncertainty over a choice of partition.

CONDITIONING SCALE AND IDENTIFIABILITY

Quarterly conditioning removes recurring hour-of-week means while retaining organisation across weeks within a quarter. Monthly conditioning absorbs more of that low-frequency organisation. In the primary raw series it lowers f_{sm} from 0.65–0.79 to 0.45–0.49 [Table III]. The result is therefore a finite-scale statement, not a partition-invariant material constant.

The 14-day sensitivity has a sharper limitation. Each (14-day block, hour-of-week) stratum contains at most two observations. After subtracting their mean, the two residuals are equal in magnitude and opposite in sign. Magnitude homogenisation then changes nothing, forcing $C_{sm} = 0$ algebraically. This row diagnoses loss of identifiability at too-fine conditioning; it cannot test whether coupling is physically absent.

PANELS, DEPENDENCE, AND CLAIM BOUNDARY

BTC, ETH, BNB, and SOL were the primary assets. XRP, ADA, DOGE, and AVAX were added after the analysis was fixed and remain a post-hoc extension. Their raw-flow point coupling shares are 0.52–0.73, but two intervals cross one half. We therefore claim a resolved majority only for the four primary raw series under quarterly conditioning.

The assets are not independent replications. Pairwise weekly-flow correlations in the primary panel are 0.14–0.39 for raw flow and −0.13–0.26 for normalised imbalance. The repeated direction across assets is a robustness pattern, while uncertainty is reported asset by asset; we do not multiply significance levels across the panel.

The original predeclared screen required raw observed variance to exceed twice the 98.75th joint-null percentile in all four primary assets. BNB failed. The exact attribution answers a different conditional question and does not amend that outcome.

TABLE I. Exact quarterly attribution. $A = V_{\mathrm{obs}}/V_0$; f_s and f_{sm} divide $V_{\mathrm{obs}} - V_0$. Intervals are paired eight-week circular moving-block intervals for f_{sm}. The first four assets are primary.

asset	flow	A	f_s	f_{sm}	95% interval	$P(f_{sm} > 1/2)$
BTC	raw	2.75	0.21	0.79	[0.63,0.88]	1.000
BTC	normalized	3.16	0.46	0.54	[0.45,0.62]	0.816
ETH	raw	3.32	0.30	0.70	[0.62,0.78]	1.000
ETH	normalized	4.24	0.41	0.59	[0.49,0.66]	0.962
BNB	raw	2.18	0.35	0.65	[0.52,0.76]	0.987
BNB	normalized	4.18	0.41	0.59	[0.53,0.65]	0.998
SOL	raw	2.53	0.35	0.65	[0.55,0.74]	0.998
SOL	normalized	2.96	0.51	0.49	[0.35,0.60]	0.430
XRP	raw	5.64	0.48	0.52	[0.43,0.60]	0.683
XRP	normalized	5.63	0.44	0.56	[0.49,0.63]	0.956
ADA	raw	2.06	0.27	0.73	[0.60,0.83]	0.997
ADA	normalized	2.82	0.46	0.54	[0.47,0.64]	0.867
DOGE	raw	3.25	0.29	0.71	[0.58,0.78]	1.000
DOGE	normalized	3.97	0.46	0.54	[0.47,0.61]	0.856
AVAX	raw	2.34	0.44	0.56	[0.40,0.64]	0.831
AVAX	normalized	3.16	0.51	0.49	[0.41,0.58]	0.436

TABLE II. Block-length sensitivity of the primary raw-flow coupling share. Each entry is the paired moving-block 95% interval.

asset	point	4 weeks	8 weeks	13 weeks	26 weeks
BTC	0.79	[0.64,0.88]	[0.63,0.88]	[0.62,0.89]	[0.60,0.90]
ETH	0.70	[0.62,0.78]	[0.62,0.78]	[0.62,0.78]	[0.62,0.78]
BNB	0.65	[0.52,0.75]	[0.52,0.76]	[0.53,0.75]	[0.56,0.74]
SOL	0.65	[0.55,0.74]	[0.55,0.74]	[0.56,0.73]	[0.59,0.72]

TABLE III. Conditioning-scale sensitivity in the primary panel. Two-point biweekly strata force equal residual magnitudes after demeaning, so their zero coupling remainder is a degeneracy, not evidence of absence.

asset	flow	conditioning	f_{sm}	95% interval	status
BTC	raw	quarterly	0.79	[0.63,0.88]	identified
BTC	raw	monthly	0.47	[0.40,0.55]	identified
BTC	raw	biweekly	0.00	[-0.00,0.00]	degenerate
BTC	normalized	quarterly	0.54	[0.45,0.62]	identified
BTC	normalized	monthly	0.49	[0.39,0.56]	identified
BTC	normalized	biweekly	-0.00	[-0.00,0.00]	degenerate
ETH	raw	quarterly	0.70	[0.62,0.78]	identified
ETH	raw	monthly	0.45	[0.37,0.53]	identified
ETH	raw	biweekly	-0.00	[-0.00,0.00]	degenerate
ETH	normalized	quarterly	0.59	[0.49,0.66]	identified
ETH	normalized	monthly	0.45	[0.36,0.51]	identified
ETH	normalized	biweekly	0.00	[-0.00,0.00]	degenerate
BNB	raw	quarterly	0.65	[0.52,0.76]	identified
BNB	raw	monthly	0.49	[0.40,0.66]	identified
BNB	raw	biweekly	0.00	[-0.00,0.00]	degenerate
BNB	normalized	quarterly	0.59	[0.53,0.65]	identified
BNB	normalized	monthly	0.45	[0.37,0.51]	identified
BNB	normalized	biweekly	0.00	[-0.00,0.00]	degenerate
SOL	raw	quarterly	0.65	[0.55,0.74]	identified
SOL	raw	monthly	0.49	[0.39,0.59]	identified
SOL	raw	biweekly	0.00	[0.00,0.00]	degenerate
SOL	normalized	quarterly	0.49	[0.35,0.60]	identified
SOL	normalized	monthly	0.41	[0.31,0.51]	identified
SOL	normalized	biweekly	0.00	[-0.00,0.00]	degenerate

TABLE IV. Legacy non-additive permutation diagnostics. R_J, R_S, and R_M divide observed variance by the medians of joint-order, sign-order, and magnitude-order nulls. The latter two break contemporaneous sign–magnitude pairing and are controls, not variance components.

asset	flow	R_J	R_S	R_M
BTC	raw	2.75	2.82	2.14
BTC	normalized	3.20	3.35	1.64
ETH	raw	3.36	3.43	2.10
ETH	normalized	4.22	4.43	1.89
BNB	raw	2.19	2.29	1.66
BNB	normalized	4.20	4.39	1.86
SOL	raw	2.55	2.61	1.75
SOL	normalized	2.99	3.14	1.52
XRP	raw	5.67	5.81	1.81
XRP	normalized	5.64	5.93	1.87
ADA	raw	2.08	2.10	1.75
ADA	normalized	2.83	2.98	1.59
DOGE	raw	3.26	3.29	2.11
DOGE	normalized	3.99	4.19	1.71
AVAX	raw	2.36	2.53	1.57
AVAX	normalized	3.18	3.35	1.55